Programming Project: Final Report

Github Link: https://github.com/LoganCaf/HouseCleanBot

1. Problem Statement

For many people, maintaining a clean & comfortable space at home is both time-consuming and mentally taxing. Sweeping, vacuuming, and mopping are the last things on many minds after a long day at work. As a result, household chores accumulate; creating stress and taking away from free time, which could otherwise be spent with friends and family, on hobbies, or simply relaxing. Additionally, unexpected messes (including pets tracking in dirt, spilled snacks, etc.) create a layer of unpredictability which may not be properly addressed by weekly cleans.

2. Solution

Our solution is a smart home-cleaning agent: powered by Deep Q-learning reinforcement learning, and a desire to create more free time for the customer. By understanding floor plans and continuously learning to navigate its environment, our agent aims to proactively manage messes by keeping track of dirty regions and finding the best way to navigate & clean the space. This transforms a tedious task into a background service, reducing stress and creating more time in the evenings and on the weekends. In our final model, it gets a reward that is 3 times larger than when taking random actions and cleans 3 times more of the map. While it can't clean all of the map, this is partially due to a high majority of the map being inaccessible, whether in small corners or behind walls that have no door access. Due to this, our final result was a cleaning percentage of about 20% on a 300 by 300 area map in 1000 moves. This shows our problem is significantly better than cleaning randomly, but with more time and resources should be improvable.

3. Demo

See demo.mp4 for a video demonstration of the model cleaning a home.

4. Assumptions, Constraints, & Implications

Our model provides the base for an incredibly adaptable cleaning agent, though there are some assumptions and limitations. First, our floor plans in the training data set consisted of single-story diagrams. While the ability to traverse stairs would likely require some hardware upgrades from current commercial options, our model would have to expand as well. Second, we are assuming that one pass through every portion of the floor means the space is clean. It would not be difficult to send the agent back through the most efficient cleaning path, but some messes may be harder to clean than others; so a real-world application of this would benefit from addressing this. Lastly, we assumed a consistent floor for training. This would likely require an additional hardware upgrade, but let's also assume this is not an issue. Our model would then have to account for variables such as carpet height and uneven surfaces. These are not the only assumptions we made in constructing our model, though they are the major constraints. Furthermore, the aforementioned limitations are not insurmountable. Although these limitations would take both time to incorporate into the model and collaboration with the engineers constructing the hardware, these limitations and constraints are the next logical steps to our current model.

5. How Solution was built

The expected input is an SVG file for a floor plan, though the model itself does not necessarily require this. There are several functions to read and analyze these SVG files before feeding them into our agent. The model itself uses Deep Q-learning with several modern enhancements that proved to be necessary for performance. These include a Double DQN architecture for training, which means the main model predicts the best action while the secondary model, which is updated way less often, predicts the q-value of that action. We also used a dueling model that separated state and action rewards, as the states in our model have a reward independent of the next action. Finally, we used a prioritized experience replay that shows the model more memories that are guessed worst on to help it train faster. After all these upgrades, the model does better than it did initially and far better than random actions do, but as always, there are always improvements to be had.

6. Summary

Our problem was to take the load off people by cleaning their homes so they can focus on more important things in their lives. To address this, we decided to use reinforcement learning, specifically deep-q-learning to clean people's homes. While this has been done in the future, I'd like to further improve speed and accuracy as well as take on more cleaning tasks than just the floor, and by having a generalizable architecture like DQN’s, this should be possible. We did, however, run into issues, including but not limited to, incredibly slow training times (nearly 1 minute per round), unstable learning, reward hacking, and a teammate leaving us without notice, causing extreme delays and many late nights.